

Nugget-o-verse

Let's break knowledge in
edible nuggets

Nugget-o-verse | For people in hurry

Definitions

Nugget

A Nugget refers to any atomic piece of consumable content, which can take various forms such as a video, an image, a question, or even an audio file. There are more than 12 types of Nuggets available. Each nugget is tagged with multiple tags, which include category, subject, chapter, topic (optional).

XP

The student earns reward points when they successfully complete a nugget or answer a question correctly. XP points are not mandatory and may not be included in every nugget. These are available in multiples of 15. *Example : 60 xp points*

Experience

A curated collection of Nuggets arranged in a specific order to serve a particular purpose. For instance, it can be designed to provide a focused learning experience for a chapter or a topic.

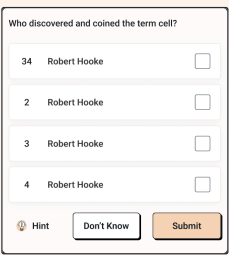
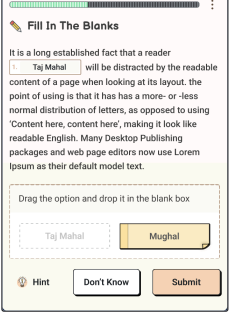
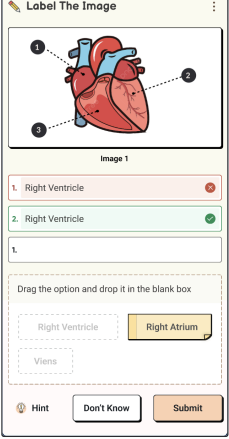
Timer

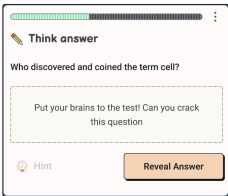
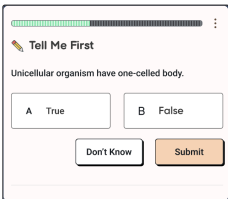
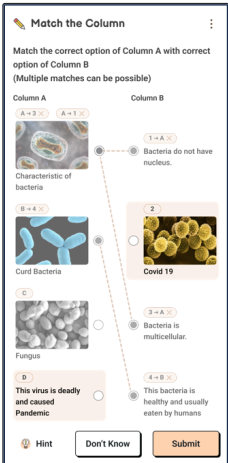
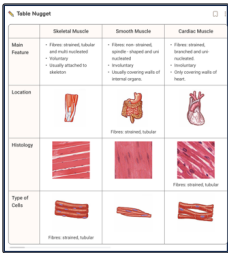
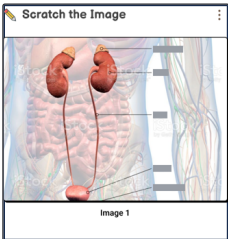
This is the maximum amount of time (in seconds) given to the student to solve a question-type nugget. Timer is available in multiples of 15. *Example : 15, 30. . .180 seconds*

Nugget variants

Nugget Type	Model	Use cases
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1	NOTE (Available in P1, P2-improved)		Combination of Heading (H1), Subheading (H2), Text, Image, UL (Unordered Lists), OL (Ordered Lists) Relevant for Summaries, Descriptions and Notes We can also highlight, bold, italic and colorize specific text segments. Make sure to keep the content of this nugget <i>CRISP</i> .
2	IMAGE (Available in P1, P2-improved)		JPEG and PNG Image types are supported. Relevant for Diagrams, Experimental setup, Reaction mechanisms, mindmaps etc The student shall spend some of their time learning from this image. Make sure it's clear and to the point.
3	VIDEO (Available in P1, P2-improved)		Secure Video URLs in <i>.mpd</i> format which are already stored in PW Admin. Relevant for video classes and clips Ideally, the videos should be short. In Phase2, a clipping feature will be available that allows you to select and save a specific segment of the video. Regardless, the videos should not exceed a duration of 1 hour.
4	AUDIO (Available in P1)		Playable audio files in mp3 format. Relevant for audio explanations, footnotes, pronunciations and mnemonics The Audio files should not be very long in length. An ideal range is between 10 seconds to 600 seconds.
5	SCC (Single choice correct Question) (Available in P1, P2-improved)		• Questions can have text, images, mathematical equations and any number of options • Hint is optional. • Solution can have any combination of text, image and mathematical equations • Only 1 correct option allowed • A timer should be added to the SCC, indicating the amount of XP points the student will receive if they answer correctly.

6	MCC (Multiple Choice Correct Question) (Available in P1, P2-improved)		• Questions can have text, images, mathematical equations and any number of options • Hint is optional. • Solution can have any combination of text, image and mathematical equations • Multiple options can be correct • A timer should be added to the MCC, indicating the amount of XP points the student will receive if they answer correctly.
7	FIB (Fill in the Blanks) (Available in P1, P2-improved)		• An FIB is a textual sentence or a paragraph with any number of blanks. Essentially it is a combination of <i>Texts</i> and <i>Blanks</i>. • Any number of options can be added in the FIB. • Hints and solutions can be added as well. • Long paragraphs can be converted to FIBs by eliminating keywords.
8	LTI (Label the Image) (Available in P1, P2-improved)		LTI are the Label the Image questions, where the basically an image based Fill in the blank. The student places chits to their respective markers Relevant for Diagrams, Experimental Setup, Reaction mechanisms.

9	SUB (Subjective Questions) (Available in P1, P2-improved)		This Nugget does not take any input from the student. The student reads the question and writes the answer in their physical notebook. They can verify the answer by clicking the “Reveal Answer” button.
10	True and False (Available in P1)		- Combination of text and image can be added to the question. Only 2 possible options (True and False) are available. - Hint and solution can also be added.
11	Match the Column (Available in P2)		- Match the Column nugget has two Columns: Column A & Column B - An element of LHC can be connected to any number of elements in RHC - Each element in both columns will have format: - only Text - only Image - Text → Image - Image → Text
12	Table Nugget (Available in P2)		Table nuggets can be used for Comparison Tables, Experimental Results, Statistical Data.
13	Scratch The Image		- Scratch the image nugget is useful to memorise and recall, Diagrams, Flow Charts and Experimental Apparatus. - The image in itself will contain the labels. - A maximum of 10 labels shall be blurred.

Use cases

1. Creation of Learning Experience with the combination of Reading and Question Nuggets

Experience

These nuggets can be plugged and played to build

- **A Test series** (a collection with a variety of Question nuggets)
- **A Video Series** (using only video nuggets) or
- **A Pedagogical Blend** (video + note + question + ...).
- **A Notes File** (with only note and image nuggets)

*Essentially, any meaningful collection of these nuggets is called an **Experience***

Why are we doing this?

Students in schools often receive education in large groups led by sub-par teachers using a one-size-fits-all approach. In a standard classroom, all students are expected to progress at the same pace. However, this approach benefits only a small fraction of students, leaving most struggling to grasp the material and resorting to memorization.

The issues with this system include:

1. **Lack of Quality Content and Teaching Methods:** The educational materials and teaching techniques employed are often of low quality.
2. **Lack of Personalized Attention:** Students don't receive individualized support, which hinders their learning experience.
3. **Absence of Self-Paced Learning:** The rigid structure of teaching doesn't accommodate varying learning speeds.

To compensate for these shortcomings and to achieve better exam results, students often resort to:

- **Enrolling in Local Tuition Centers:** These centers offer better content and teaching methods compared to schools but still fall short in overall quality. Additionally, their fixed schedules hinder self-paced learning.
- **Using Online Resources:** Students search for educational content on platforms like YouTube, but finding relevant videos aligned with their curriculum is challenging and requires self-motivation. Even if suitable videos are found, they lack interactive practice and guidance.
- **Experimenting with Learning Apps:** Some students try out multiple learning apps from app stores, which can be exciting, but the abundance of choices can also be overwhelming.

In essence, the current school system's group-based, uniform approach to teaching leads to various challenges. Students often resort to external resources and methods to fill the gaps in their learning and improve their exam performance.

So, what's the Problem and Solution?

Mobile apps digitize educational content (videos, questions and pdfs) and offer it on a freemium model.

The learning experience largely remains the same across apps with the content having some variety. Basically, these apps act as digital libraries of curated educational content. This mode solves the problem of high-quality curated content, but the problem of personalized learning and handholding remains unsolved.

The CRUX of the matter in self-paced education is NOT the mere existence of high-quality content, but rather, its delivery in a captivating manner that can sustain learners' interest for extended durations, thereby facilitating optimal learning outcomes.

In a world where students' attention spans are alarmingly brief, where an overflow of free content (available on platforms such as Telegram, YouTube, and websites) is frequently pirated, and where retention rates are dismally low, the dire necessity for a pedagogy-driven product is evident.

Also, content may be pirated but not the Pedagogy 🧩

✔ **Nugget-o-verse lays the foundation of building Pedagogy-driven products at PW**